Chong Chan How

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EDUCATION

National University of Singapore **Graduating 05/2026**
Bachelor of Computing in Computer Science **GPA 4.51 / 5.00**

EXPERIENCES

On-Premises Infrastructure Intern – DSTA **01/2025 – 07/2025**

- Led an exploratory project to assess the feasibility of adopting a Type-I hypervisor for enterprise infrastructure, evaluating performance, high availability, disaster recovery, and vendor ecosystem support.
- Built a simulated air-gapped on-premises environment and automated the deployment of core infrastructure services (DNS/DHCP, package repositories, image registry) using Ansible and OpenShift, improving provisioning repeatability and reliability.

Code Analysis Research Assistant – NUS **08/2024 – 08/2025**

- Performed large-scale analysis of Python code submissions to identify recurring error patterns and developed a classification framework to categorize issues by type and logical causes.
- Contributed analysis report used by faculty to evaluate student learning challenges and refine course curriculum.
- Implemented an automated marking script in Python to batch-test and evaluate hundreds of code submissions, improving analysis efficiency and reducing project timeline by 1 month.

SKILLS

- **Infrastructure & Systems:** Linux administration, Type-I Hypervisors (Proxmox), virtualization, Docker containers, Kubernetes orchestration, on-premises & hybrid infrastructure.
- **Network Architecture & Configuration:** DNS, DHCP, IP subnetting, VLANs, SDN concepts
- **Cloud & Platform Tools:** OCI (Oracle Cloud Infrastructure), Git & version control, container registries, package repositories, Redis Cloud
- **Security & Reliability:** Disaster recovery planning, high availability concepts, air-gapped environment deployment, Role-Based Access Control (RBAC).
- **Automation & DevOps**: Ansible, Python, Bash scripting, CI/CD
- **Programming:** Python, Java, JavaScript, C++, P4

PROJECTS

PeerPrep Platform

- A full-stack microservice-based web application for peer-matching students to practice LeetCode-style technical interview questions in real-time.
- Developed with **React** (frontend), **Express** (backend), and **MongoDB** & **Google Cloud** (data storage).
- Implemented **RESTful APIs** for seamless communication between services.
- Link to Repository: https://github.com/CS3219-AY2425S1/cs3219-ay2425s1-project-g34

Home Lab Environment

- Designed and deployed a **hybrid self-hosted infrastructure** using Proxmox, hosting virtual machines and LXC containers alongside cloud services to achieve an **accessible** and **cost-efficient** setup.
- Configured additional internal services including local DNS, automation scripts (e.g., scheduled email notifications, web scrapers), and a centralized dashboard for **monitoring** and **control**.
- Focused on hybrid infrastructure management, cost optimization, and secure network configuration, applying principles of automation, scalability, and system reliability.